Lecture 17 // Oct. 20 // Proof of the NST theorem

Last time: We defined the r_s-invariant for $s \in [-\infty, 0]$

$$r_s(Y) = -\inf_{r<0} \{ r \mid \exists\ \Theta\text{-supported } [r, -s] \text{ cycle} \}$$

which asks the question "In which filtered subquotients do we have a cycle of the form $\Theta^t \dots$"

1) r_s-invt. is a piecewise const. non-inc. fxn

2) If $\exists$ negative-def. cob. W (with $H_1(W)=0$) from Y_1 to Y_2, then
$r_s(Y_1) \leq r_s(Y_2)$
In Part., r_s is a hom. cob. inv.: If Y bounds a negative-def. 4-mfd W (with $H_1(W)=0$), then $r_s(Y) = \infty$

3) In addl.,
If, in addl., $\pi_1(W) = 1$, then the inequality is strict (allowing $\infty < \infty$)

Today, we'll prove one of the main thms of the NST paper.

Thm [NST] Let $K \subset S^3$. Denote $Y_n = S^3_{1/n}(K)$, $n > 0$.
If $[D_2^{Y_1}] \neq 0$, then none of the Y_n are hx-cob to eachother.

If $r_s(Y_1) < \infty$ for any s, then $\{Y_n\}_{n=1}^{\infty}$ is lin. indep. in $\Theta^3_{\mathbb{Z}}$

Rmk A-priori, it's not obvious that $\Theta^3_{\mathbb{Z}}$ is at all interesting. For many years after def, not known whether $\Theta^3_{\mathbb{Z}}$ is 0, or ∞-gen. It is: In particular, it contains a $\mathbb{Z}^\infty$ subgroup. ($\bigoplus_{i=1}^{\infty} \mathbb{Z}_i$)
The o.g. proof was due to Fintushel and FS in the 80's.
In some sense, this thm is a generalization of theirs.
FFS had same family, and also used Yang-Mills theory.
(Modern language (Floer) allows them to generalize it)

Rmk It's actually very difficult to study $1/n$ surgeries using HF. There's no known inv. coming from HF which distinguish ∞-ly many $1/n$ surgeries on a fixed knot.

To study lin. combos of Y_n, we must first understand the behav. of r_s under #.

Thm For any $s \in [-\infty, 0]$ and $s_1, s_2 \in [-\infty, 0]$ such that $s_1 + s_2 = s$,

$$r_s(Y_1 \# Y_2) \geq \min\{ r_{s_1}(Y_1) + s_2,\ r_{s_2}(Y_2) + s_1 \}$$

Perhaps the form of this theorem – ie, it is an inequality, and takes a minimum over a sum of r_s and s's – is strange. We'll first give a pseudo-pf of the inequality, since the reason why why it's of such a strange form will become apparent during the proof

Pf (sketch)
We'll pretend that $\underline{C}(Y_1 \# Y_2) = \underline{C}(Y_1) \otimes \underline{C}(Y_2)$
This ($\#$ corresponds to $\otimes$) is the behav. one usually expects for complexes under connect sum.

The reducible in $\underline{C}(Y_1 \# Y_2)$ is $\Theta_1 \otimes \Theta_2$, and
$\widetilde{CS}(\alpha \otimes \beta) = \widetilde{CS}(\alpha) + \widetilde{CS}(\beta)$.

Now, fix s_1 and s_2. Spse that we have a ΘSC α for $\underline{C}(Y_1)$ in $[r_1, -s_1]$, and a ΘSC β for $\underline{C}(Y_2)$ in $[r_2, -s_2]$.

What "Θ-supported" means is that in $\underline{C}(Y_1)$, $\alpha = \Theta + \dots$.
To say that α is a cycle in this subqu. is to say
that a) α is in the subquotient, ie, $\widetilde{CS}(\alpha) \leq -s$
and b) $\widetilde{CS}(\underline{\partial}\alpha) < r_1$. In particular, $\partial\alpha$ need not be 0,
but if it isn't, it's outside subquotient.
The same is true for β: $\widetilde{CS}(\beta) \leq -s_2$ and $\widetilde{CS}(\underline{\partial}\beta) < r_2$.

Claim: $\alpha \otimes \beta$ is a $\Theta_1 \otimes \Theta_2$-supported
(a.k.a., Θ-supp in the $\otimes$)
we know that $\widetilde{CS}(\alpha \otimes \beta) \leq (-s_1) + (-s_2) = -s$.
So $\alpha \otimes \beta$ is in the subquotient.
By the Leibniz rule,
$$\underline{\partial}(\alpha \otimes \beta) = (\underline{\partial}\alpha) \otimes \beta + \alpha \otimes (\underline{\partial}\beta).$$
Individually, $(\underline{\partial}\alpha) \otimes \beta < r_1 + |-s_2|$ and
$\alpha \otimes (\underline{\partial}\beta) < (-s_1) + r_2$
Therefore $\alpha \otimes \beta$ is a cycle in the
subquotient $\left[\max\{r_1 + (-s_2), (-s_1) + r_2, -s\}\right]$.

This is where the formula appears.

Now, note that this is true $\forall$ ΘSC in these subqt's.
When does this occur? By def^n, $-r_3(Y_1)$ and

$-r_{s_2}(Y_2)$ are the borders of when $\exists$ ΘSC in each resp. subqt. So the claim is that you can take any r_1 up to $-r_{s_1}(Y_1)$ and any r_2 up to $r_2 \longrightarrow -r_{s_2}(Y_2)$.

Thus we get a ΘSC in the subq's

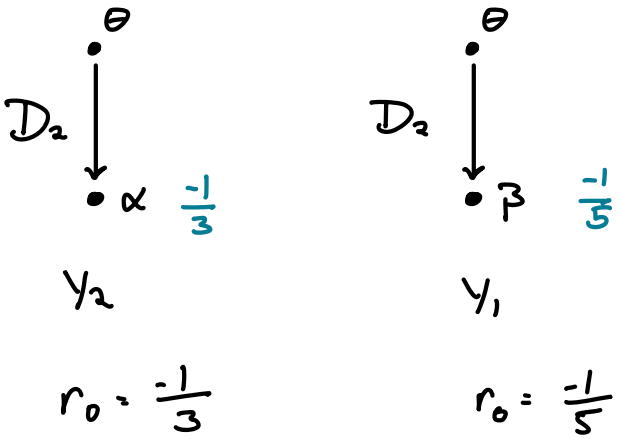
$$\left[\max\left\{-r_{s_1}(Y_1)+(-s_2),\ -r_{s_2}(Y_2)+(-s_1)\right\},\ -s\right]$$

$$\Rightarrow\ r_s(Y_1\#Y_2) \geq \max\left\{r_{s_1}(Y_1)+s_2,\ r_{s_2}(Y_2)+s_1\right\}\ \square$$

Cor $\quad r_0(Y_1\#Y_2) \geq \min\{r_0(Y_1), r_0(Y_2)\}$

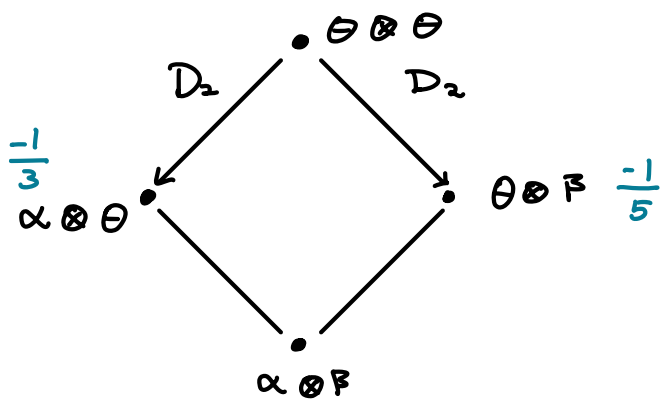
To see why this behavior — taking a minimum instead of adding — is more natural in this setting, lets do an example.

Example Let the following be the u-bar cpxes for Y_1 and Y_2.



Here, r_0 is measuring the cs displacement of the D_2 arrow.

The point is that if we pretend $\#$ is $\otimes$, the Complex for $Y_1\#Y_2$ looks like:



Notice that $D_2(\theta)$ has 2 arrows, corr to the 2 summands in the Leibniz rule.

Here, it's clear that to kill all the D_2's, you must take the minima of all of the summands.

Unfortunately, it's not acc. true, in general, that

$$\underline{C}(Y_1 \# Y_2) = \underline{C}(Y_1) \otimes \underline{C}(Y_2).$$

The rough reason is that it turns out that $\underline{C}(Y_1 \# Y_2)$ cannot be determined from $\underline{C}(Y_1)$ and $\underline{C}(Y_2)$ alone. To see why, lets look at the potential types of generators of $\underline{C}(Y_1 \# Y_2)$:

- $\theta \otimes \theta$ (red ⊗ red) } This is red
- $\alpha \otimes \beta$ (irred ⊗ irred)
- $\theta \otimes \beta$ (red ⊗ irred) } These are irred in the tensor product
- $\alpha \otimes \theta$ (irred ⊗ red)

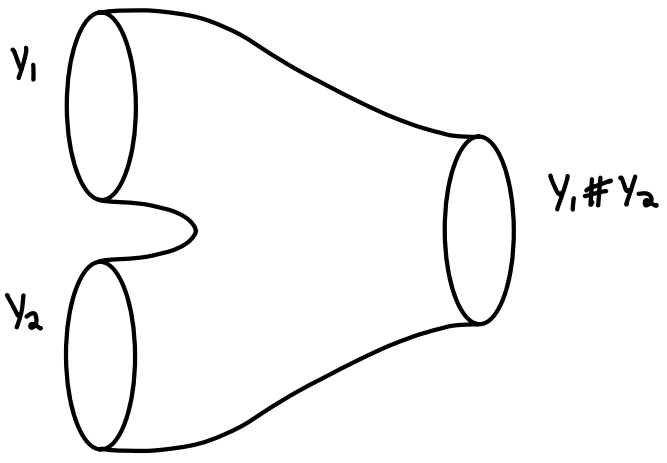
The reason why the last two, which are combinations of reducible and irreducible generators, are irreducible in the product is as follows. Recall that generators correspond to $SU(2)$ representations, and that $\pi_1(Y_1 \# Y_2) \cong \pi_1(Y_1) * \pi_1(Y_2)$. If we have $SU(2)$ reps. ρ_1 and ρ_2 of $\pi_1(Y_1)$ and $\pi_1(Y_2)$, and ρ_1 has trivial centralizer but ρ_2 doesn't, then $\rho_1 * \rho_2$ also has trivial centralizer. The intuition here is the same as for gauge transformations on the composite cylinder: even if some element commutes with ρ_2,

It doesn't with ρ_1, so it doesn't commute with $\rho_1 * \rho_2$ either.

The issue arises because $\alpha \otimes \Theta$ and $\Theta \otimes \beta$ are irreducible: there is no reason to expect that there are no flows into/between them! In particular, we should be able to have maps from $\alpha \otimes \beta$ to $\alpha \otimes \Theta$. However, even if there is a D_1 map where $D_1(\beta) = \Theta$, in choosing the u-bar cpx, we threw these out! So you actually need both the D_1 and D_2 maps of each factor to calculate the u-bar cpx of the connect sum.

Rmk Being able to write down an explicit formula for $\underline{C}(Y_1 \# Y_2)$ as some tensor product is one motivation for why one may want to try harder to incorporate both D_1 and D_2 into the diff'l.

What salvages this is the fact that there is a cobordism map $\underline{C}(Y_1) \otimes \underline{C}(Y_2) \to \underline{C}(Y_1 \# Y_2)$. This cobordism is literally the pair of pants, from $Y_1 \sqcup Y_2$ to $Y_1 \# Y_2$.



Thus far, we've assumed that the Y's are connected which seems silly to note, but the distinction btwn conn and disconn. boundary really matters to the gauge gp: they're qualitatively different. It turns out that we get get bordism maps from disconn $\to$ conn, but not the other way. But this means that the rs inv. of the algebraic cpx $\underline{C}(Y_1) \otimes \underline{C}(Y_2)$ bounds the rs inv. of the complex we actually care about, $\underline{C}(Y_1 \# Y_2)$.

Q Is the inequality always strict?

A No! For example, take $Y_1 = -\Sigma(2,3,5)$ and $Y_2 = \Sigma(2,3,5)$. Then $Y_1 \# Y_2$ is homology cobordant to S^3. The U-bar complex for each summand looks like



$r_0(Y_1) = \frac{-1}{120}$ $\qquad$ $r_0(Y_2) = \infty$

Since rs is a hx cob inv.,

$$\infty = r_0(S^3) = r_0(Y \# -Y) \geq \min\{r_0(Y), r_0(-Y)\}$$

$$\geq \min\left\{\frac{1}{120}, \infty\right\}$$

$$\geq \frac{1}{120}$$

We are now prepared to prove the thm. We'll prove a more general algebraic statement.

Thm Let $\{Y_n\}_{n=1}^{\infty}$ be a sequence of $\mathbb{Z}HS^3$'s such that

1) $\infty > r_0(Y_1) > r_0(Y_2) > \dots$

2) $\infty = r_0(-Y_1) = r_0(-Y_2) = \dots$

Then the $\{Y_n\}_{n=1}^{\infty}$ are lin. indep. in $\Theta^3_{\mathbb{Z}}$.

Pf Assume not: spse $\sum_{i\le m} n_i Y_i = 0$ in $\Theta^3_{\mathbb{Z}}$.

Then $n_m Y_m = \sum_{i<m} n_i Y_i$ (assume $n_m > 0$),

so $r_0(n_m Y_m) = r_0\left(\sum_{i<m} n_i Y_i\right)$. Note that these n_i can be positive or negative

Iterating this inequality, we get that

$r_0(n_m Y_m) = \dots \ge \min\{ r_0(\pm Y_1), r_0(\pm Y_2), \dots \}$

where the $+$ or $-$ in each $\pm$ is chosen based on $\operatorname{sign}(n_i)$.

OTOH, Since $Y_m = n_m Y_m \# (-n_{m-1}) Y_m$,

$$r_0(Y_m) = r_0\left(n_m Y_m \# [-n_{m-1}) Y_m\right)$$

$$\ge \min\{ r_0(n_m Y_m), r_0(-Y_m) \}$$
$$\ge \min\{ r_0(n_m Y_m), \infty \}$$
$$= r_0(n_m Y_m)$$

since $r_0(-Y_m) = 0$ by hypothesis.

Thus $r_0(Y_m) \ge r_0(n_m Y_m)$, so

$$r_0(Y_m) \ge \min\{ r_0(\pm Y_1), r_0(\pm Y_2), \dots \}$$

which contradicts the initial hypotheses. □

The approx. intuition for this result is that small and nonzero CS displacement of D_2 cannot lie in the span of large CS displacemat.

Does this apply to our situation?
Yes! Heres a slick topological argument, which concludes the proof of [NST].

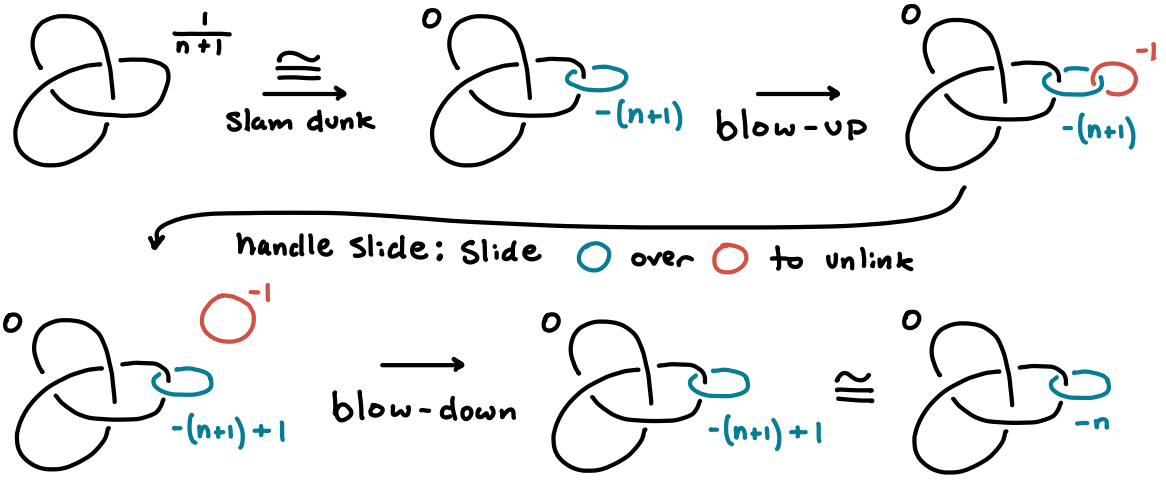
Thm [NST] Let $K \subseteq S^3$. Denote $Y_n = S^3_{1/n}(K)$, $n > 0$.
If $[D_2^{Y_1}] \neq 0$, then none of the Y_n are hx-cob to eachother.

Pf We only need to show that the conditions of the prev. thm are satisfied.
Eg, we need to verify that

1) $\infty > r_0(Y_1) > r_0(Y_2) > \dots$
2) $\infty = r_0(-Y_1) = r_0(-Y_2) = \dots$

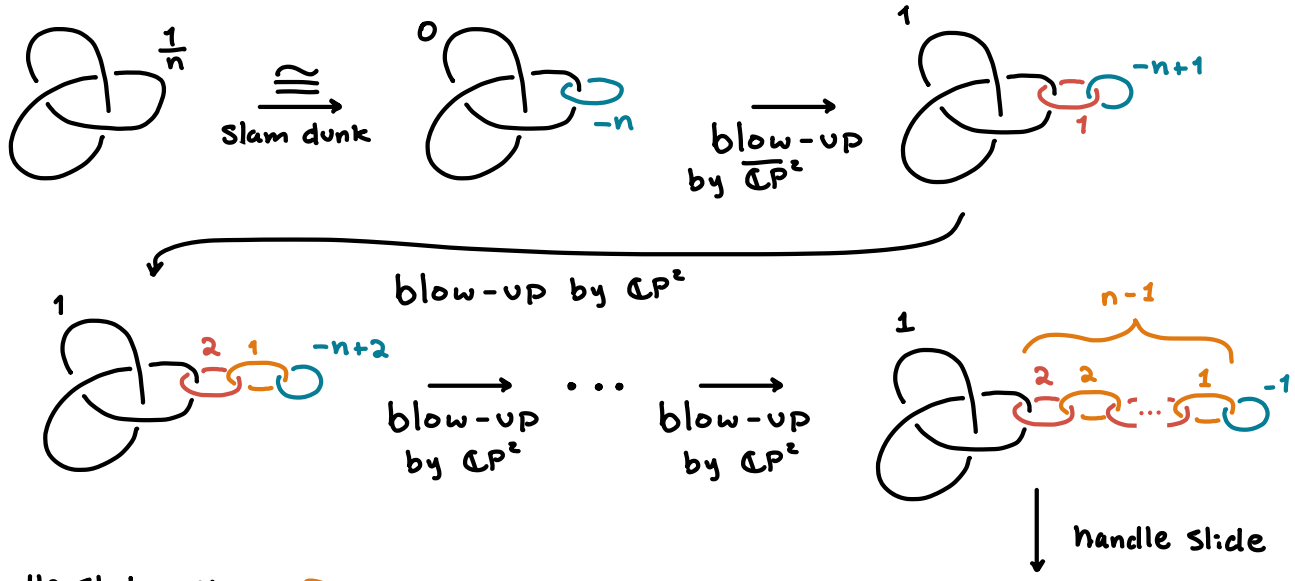
1) holds from last time:

We constructed a $\pi_1 = 1$ neg.-def. cob from Y_{n+1} to Y_n:



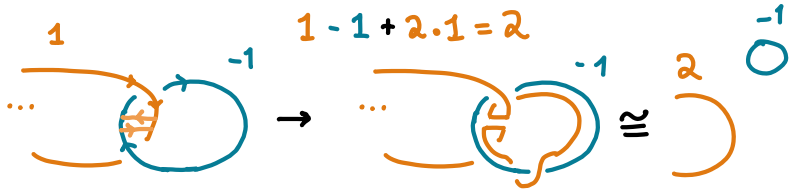
2) Follows from the fact that $S^3_{1/n}(K)$ bounds a +-def mfd if $n>0$ and a --def mfd if $n<0$.

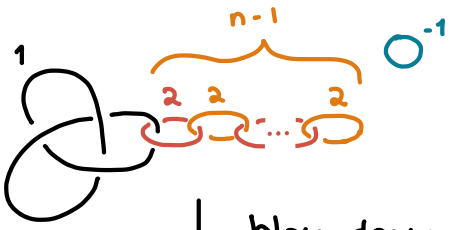
The +-def mfd can be constructed as follows ($n>0$):



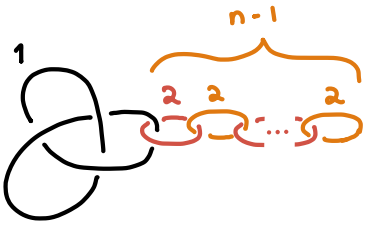
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blow down



The intersection form is $\begin{matrix} 1\{ \\ n-1\{ \\ 1\{ \end{matrix}\begin{bmatrix} 1 & 1 & & & & \\ 1 & 2 & 1 & & 0 & \\ & 1 & 2 & 1 & & \\ & & 1 & \ddots & & \\ & 0 & & & 2 & 1 \\ & & & & 1 & 2 \end{bmatrix}$, which we leave to the reader to verify has Smith normal form Id_4.

The same process can be applied when $n<0$ by blowing up with $\overline{\mathbb{C}P^2}$'s.

Since this process gives a neg-def. cob from $-Y_n$ to S^3, we can conclude that $r_0(-Y_n)=\infty$.